

AWS – Building a Virtual Private Cloud(VPC)

Creating a VPC workflow:

VPC

>

Your VPCs

>

Create VPC

>

Create VPC resources

Create VPC workflow

Success

Details

✔ Create VPC: [vpc-03e8f6fc7894efe0a](#)

✔ Enable DNS hostnames

✔ Enable DNS resolution

✔ Verifying VPC creation: [vpc-03e8f6fc7894efe0a](#)

✔ Create subnet: [subnet-0ed2a474ef25d2d56](#)

✔ Create subnet: [subnet-0fdb61579970e661f](#)

✔ Create Internet gateway: [igw-0ef51410ef7040a59](#)

✔ Attach internet gateway to the VPC

✔ Create route table: [rtb-09ac297532d2c431d](#)

✔ Create route

✔ Associate route table

✔ Allocate elastic IP: [eipalloc-08bfefacd0f66e640](#)

✔ Create NAT gateway: [nat-014946dea53a6699e](#)

✔ Wait for NAT Gateways to activate

✔ Create route table: [rtb-07850f90da649644a](#)

✔ Create route

✔ Associate route table

✔ Verifying route table creation

Related VPC resources and services

Discover additional resources you can launch within your VPC to build your network architecture. From security components to connectivity services, explore the building blocks available for your infrastructure.

Client VPN

Networking

AWS Client VPN is a managed client-based VPN service that enables you to securely access AWS resources and resources in your on-premises network.

Create

Resources

Let us know what you would like to create in your VPC

e.g., Instances, Network ACL, Transit gateway...

100 characters available. Do not disclose any personal, commercially sensitive, or confidential information.

CloudShell

Feedback

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Creating a public subnet:

The screenshot displays the AWS Management Console interface, specifically the Subnets and Route tables sections. The top section shows the Subnets page with a notification that a new subnet, 'subnet-0296b7b2ad4bd8ffa', has been successfully created. Below this, a table lists the subnets, including 'lab-subnet-public2' with ID 'subnet-0296b7b2ad4bd8ffa', which is in an 'Available' state and associated with VPC 'vpc-03e8f6fc7894efe0a'. The bottom section shows the Route tables page, listing several route tables. The 'lab-rtb-public' route table (ID: 'rtb-09ac297532d2c431d') is selected, and its 'Subnet associations' tab is active. This tab shows two explicit subnet associations: 'lab-subnet-public1-us-east-1a' (ID: 'subnet-0ed2a474ef25d2d56') and 'lab-subnet-public2' (ID: 'subnet-0296b7b2ad4bd8ffa'). The bottom section also shows a section for 'Subnets without explicit associations' which is currently empty.

Subnets (1)

You have successfully created 1 subnet: subnet-0296b7b2ad4bd8ffa

Find subnets by attribute or tag

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
lab-subnet-public2	subnet-0296b7b2ad4bd8ffa	Available	vpc-03e8f6fc7894efe0a lab-vpc	Off	10.0.2.0/24	-

Route tables (1/6)

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
lab-rtb-private1-us-east-1a	rtb-07850f90da649644a	2 subnets	-	No	vpc-03e8f6fc7894efe0a lab-vpc	219829739033
-	rtb-0af233ee4173116b6	-	-	Yes	vpc-03e8f6fc7894efe0a lab-vpc	219829739033
-	rtb-004d3001a3a609ff9	-	-	Yes	vpc-008b7097f01833b47	219829739033
Work Public Route Table	rtb-0b234327f1479c80c	subnet-0af05827897fa4e...	-	No	vpc-08d23d5ed1e22e44a Wor...	219829739033
lab-rtb-public	rtb-09ac297532d2c431d	2 subnets	-	No	vpc-03e8f6fc7894efe0a lab-vpc	219829739033
-	rtb-0b838e4ad8f9d5d91	-	-	Yes	vpc-08d23d5ed1e22e44a Wor...	219829739033

rtb-09ac297532d2c431d / lab-rtb-public

Details | Routes | **Subnet associations** | Edge associations | Route propagation | Tags

Explicit subnet associations (2)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
lab-subnet-public1-us-east-1a	subnet-0ed2a474ef25d2d56	10.0.0.0/24	-
lab-subnet-public2	subnet-0296b7b2ad4bd8ffa	10.0.2.0/24	-

Subnets without explicit associations (0)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

No subnets without explicit associations

Resource Map:

VPC dashboard < AWS Global View Filter by VPC

Your VPCs VPC encryption controls

Your VPCs (1/3) Info Last updated 2 minutes ago Actions Create VPC

Find VPCs by attribute or tag

Name	VPC ID	State	Encryption ...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR
-	vpc-008b7097f01833b47	Available	-	-	Off	172.31.0.0/16	-
Work VPC	vpc-08d23d5ed1e22e44a	Available	-	-	Off	10.0.0.0/16	-
lab-vpc	vpc-03e8f6fc7894efe0a	Available	-	-	Off	10.0.0.0/16	-

vpc-03e8f6fc7894efe0a / lab-vpc

Resource map Info Show all details

VPC Your AWS virtual network

lab-vpc

Subnets (4) Subnets within this VPC

- us-east-1a
 - lab-subnet-public1-us-east-1a
 - lab-subnet-private1-us-east-1a
- us-east-1b
 - lab-subnet-public2
 - lab-subnet-private2

Route tables (3) Route network traffic to resources

- lab-rtb-private1-us-east-1a
- rtb-0af233ee4173116b6
- lab-rtb-public

Network Connections (2) Connections to other networks

- lab-igw
- lab-nat-public1-us-east-1a

Creating a Security group:

VPC dashboard < AWS Global View Filter by VPC

Security Groups sg-0e8d6753d79ff3770 - Web Security Group

Security group (sg-0e8d6753d79ff3770 | Web Security Group) was created successfully Details

sg-0e8d6753d79ff3770 - Web Security Group Actions

Details

Security group name Web Security Group	Security group ID sg-0e8d6753d79ff3770	Description Enable HTTP access	VPC ID vpc-03e8f6fc7894efe0a
Owner 219829739033	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules Outbound rules Sharing VPC associations Related resources - new Tags

Inbound rules (1) Manage tags Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-098e3740e8f54eb61	IPv4	HTTP	TCP	80	0.0.0.0/0	Permit web requests

Launch Instance:

Success
Successfully initiated launch of instance `i-0f8b3c185196c264b`

► **Launch log**

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing and free tier usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.
[Create billing alerts](#)

Connect to your instance
Once your instance is running, log into it from your local computer.
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)
[Create a new RDS database](#)
[Learn more](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots.
[Create EBS snapshot policy](#)

Manage detailed monitoring
Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.
[Manage detailed monitoring](#)

Create Load Balancer
Create an application, network gateway or classic Elastic Load Balancer.
[Create Load Balancer](#)

Create AWS budget
AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.
[Create AWS budget](#)

Manage CloudWatch alarms
Create or update Amazon CloudWatch alarms for the instance.
[Manage CloudWatch alarms](#)

Disaster recovery for your instances
Recover the instances you just launched into a different Availability Zone or a different Region using AWS Elastic Disaster Recovery (DRS).
[Disaster recovery for your instances](#)

Monitor for suspicious runtime activities
Amazon GuardDuty enables you to continuously monitor for malicious runtime activity and unauthorized behavior, with near real-time visibility into on-host activities occurring across your Amazon EC2 workloads.
[Monitor for suspicious runtime activities](#)

Get instance screenshot
Capture a screenshot from the instance and view it as an image. This is useful for troubleshooting an unreachable instance.
[Get instance screenshot](#)

Get system log
View the instance's system log to troubleshoot issues.
[Get system log](#)

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Private subnet:

Success
You have successfully created 1 subnet: `subnet-0b5a0ec3f50cb4fed`

Subnets (1) Info
Find subnets by attribute or tag
Subnet ID: `subnet-0b5a0ec3f50cb4fed` [Clear filters](#)

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 Gateway
lab-subnet-private2	subnet-0b5a0ec3f50cb4fed	Available	vpc-03e8f6fc7894efe0a lab-vpc	Off	10.0.3.0/24	-	-

Select a subnet

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EC2 > Instances

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Instances (2)

Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
☐	Web Server 1	i-0f8b3c185196c264b	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1b	ec2-34-239-251-14.co...	34.239.251.14	-
☐	Bastion Host	i-064495fcb7b18f0f	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-3-94-6-56.compute...	3.94.6.56	-

Select an instance

CloudShellFeedback

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Web Server:

←↻⚠️ Not secure 34.239.251.14

aws

Load Test

RDS

Meta-Data	Value
InstanceId	i-0f8b3c185196c264b
Availability Zone	us-east-1b

Current CPU Load: 0%

Report:

AWS

00:08

▶ Start Lab

■ End Lab

🔍 AWS Details

📄 Details

✕

Submit

Submission Report

Grades

Task 4: Launch a Web Server Instance

In this task, you will launch an Amazon EC2 instance into the new VPC. You will configure the instance to act as a web server.

33. In the search box to the right of **Services**, search for and choose **EC2** to open the EC2 console.

34. From the **Launch Instance** menu choose **Launch instance**.

35. Name the instance:

- Give it the name **web Server 1**

When you name your instance, AWS creates a tag and associates it with the instance. A tag is a key value pair. The key for this pair is ***Name***, and the value is the name you enter for your EC2 instance.

36. Choose an AMI from which to create the instance:

- In the list of available *Quick Start* AMIs, keep the default **Amazon Linux** selected.
- Also keep the default **Amazon Linux 2023 AMI** selected.

The type of *Amazon Machine Image (AMI)* you choose determines the Operating System that will run on the EC2 instance that you launch.

Total score	25/30
Task 1 - VPC created correctly	5/5
Task 2a - New subnets created correctly	5/5
Task 2b - Subnet route table association	5/5
Task 3 - Security group created correctly	5/5
Task 4a - EC2 instance created correctly	0/5
Task 4b - EC2 instance website accessible	5/5